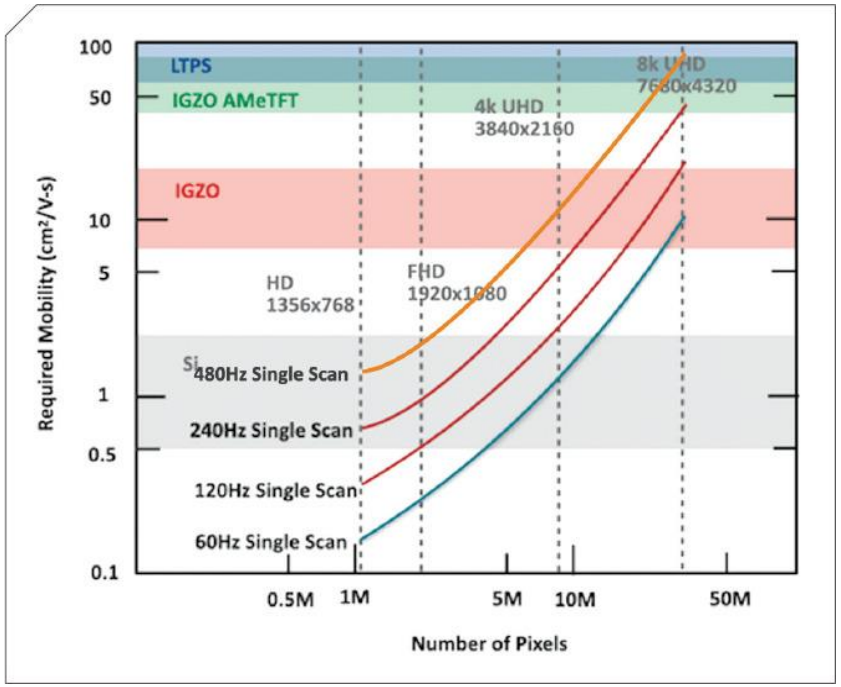
The heteroepitaxial growth of indium oxide thin film transistor with passivation layer

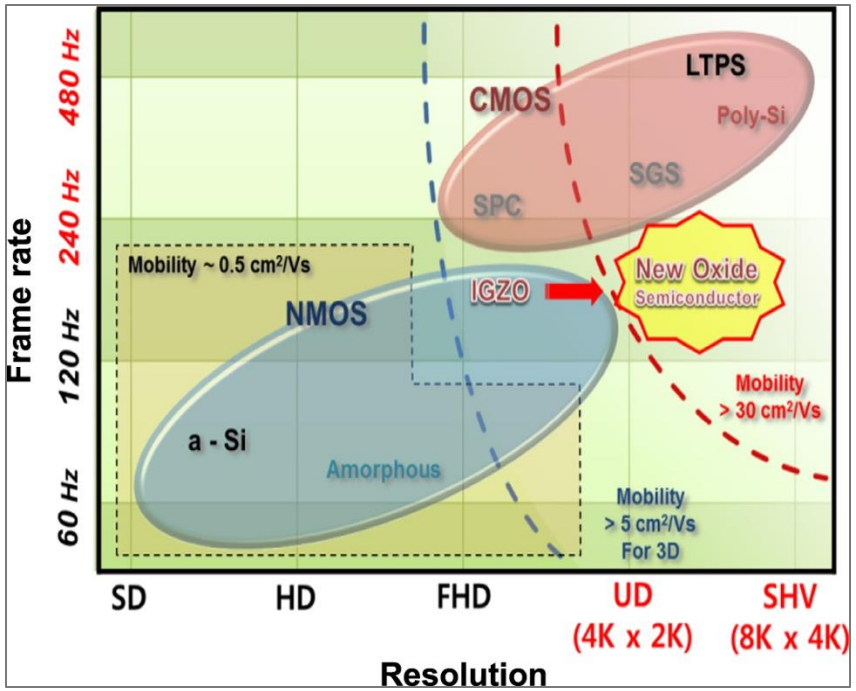
Park Jin Hong
Kyungpook National University

Demand for higher mobility of oxide TFT

Benefits of the higher mobility and trade-off with operating stress performance

- High resolution(≥200 ppi) and high frequency frame(≥240 Hz) driving have become a megatrend for the next generation active-matrix(AM) displays, which enable a vivid image and a natural motion picture in the resulting display.
- Higher TFT mobility enables smaller transistors with greater current capability. Moreover, switching speed increases as a function of mobility.



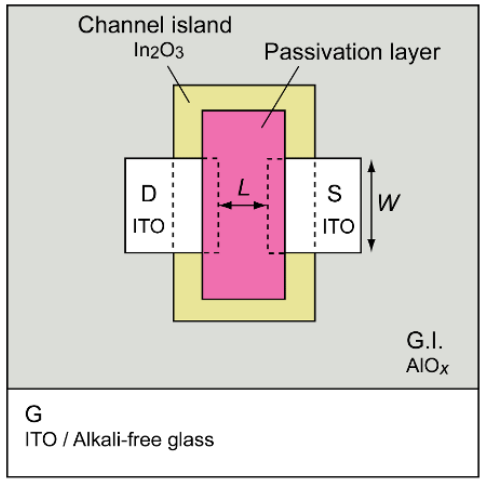


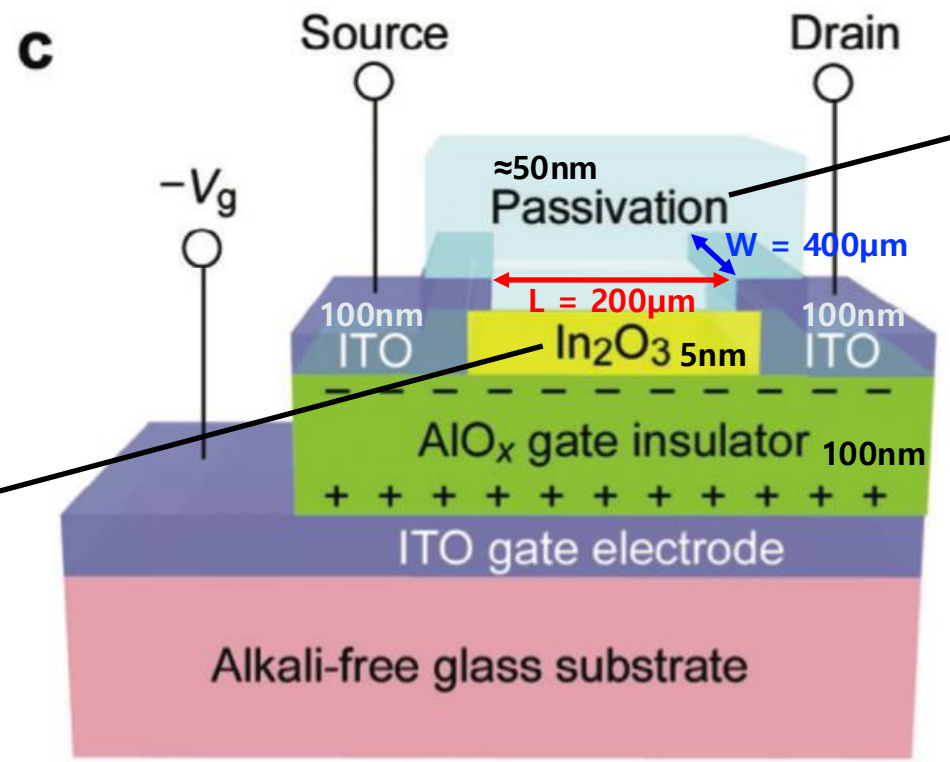
Information Display 2022,38,60.
Semicond. Sci. Technol. 2015,30,024002.

Overcome In_2O_3 TFT reliability issue

Schematic illustration of an In_2O_3 TFT with a various surface passivation layer.

- Ionic radius of the Ln^{3+} is close to that of In^{3+} and the crystal structure of Ln_2O_3 is similar to that of In_2O_3





Annealed at 300℃ for 0.5h
to crystallize In_2O_3 active layer

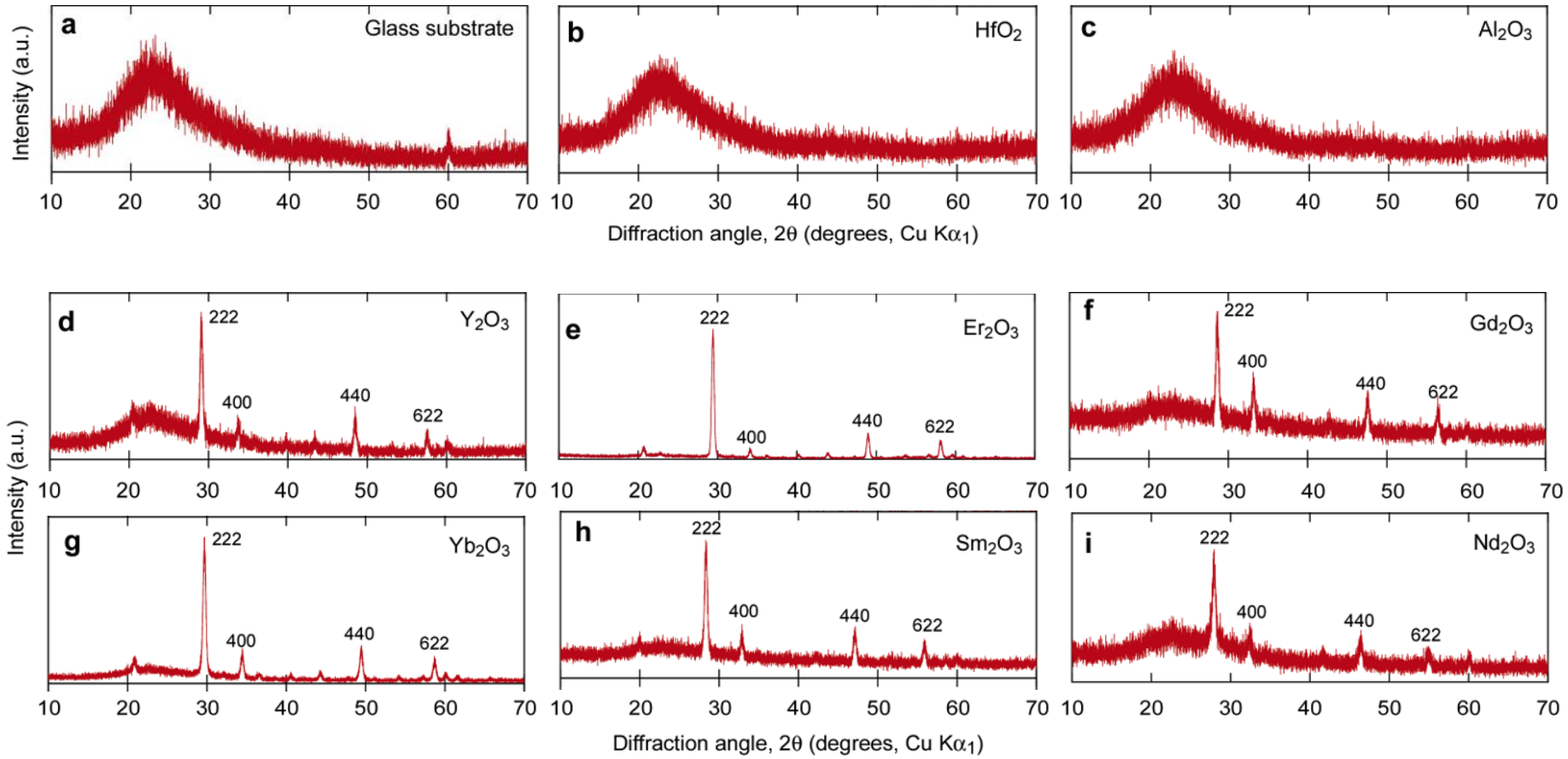
Passivation layers(annealed at 370℃ for 0.5h)
- w/o passivation
- Amorphous HfO_2
- Amorphous Al_2O3
- Polycrystalline Y_2O3(C-rare earth structure)
- Polycrystalline Er_2O_3(C-rare earth structure)
- Polycrystalline Gd_2O_3(C-rare earth structure)
- Polycrystalline Yb_2O_3(C-rare earth structure)
- Polycrystalline Sm_2O_3(B-rare earth structure)
- Polycrystalline Nd_2O_3(A-rare earth structure)

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Overcome In_2O_3 TFT reliability issue

XRD patterns of the passivation layers after thermal annealing at 370°C

- HfO_2, Al_2O_3 thin film → amorphous
- Ln_2O_3(Y_2O_3, Er_2O_3, Gd_2O_3, Yb_2O_3, Sm_2O_3, Nd_2O_3) thin film → polycrystalline

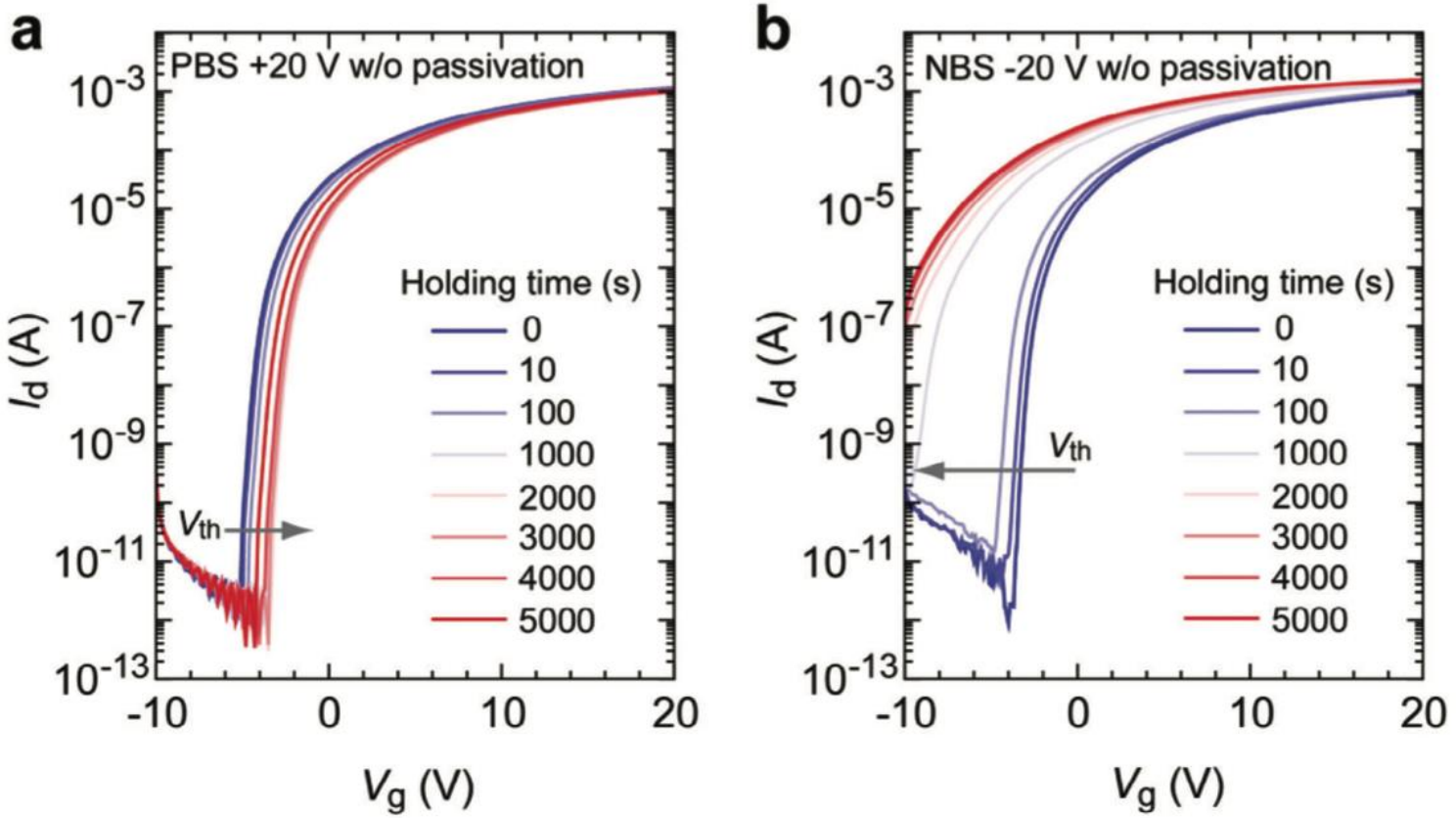


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The In_2O_3 TFTs without a passivation layer

Transistor characteristics of the In_2O_3 TFTs after applying bias stress

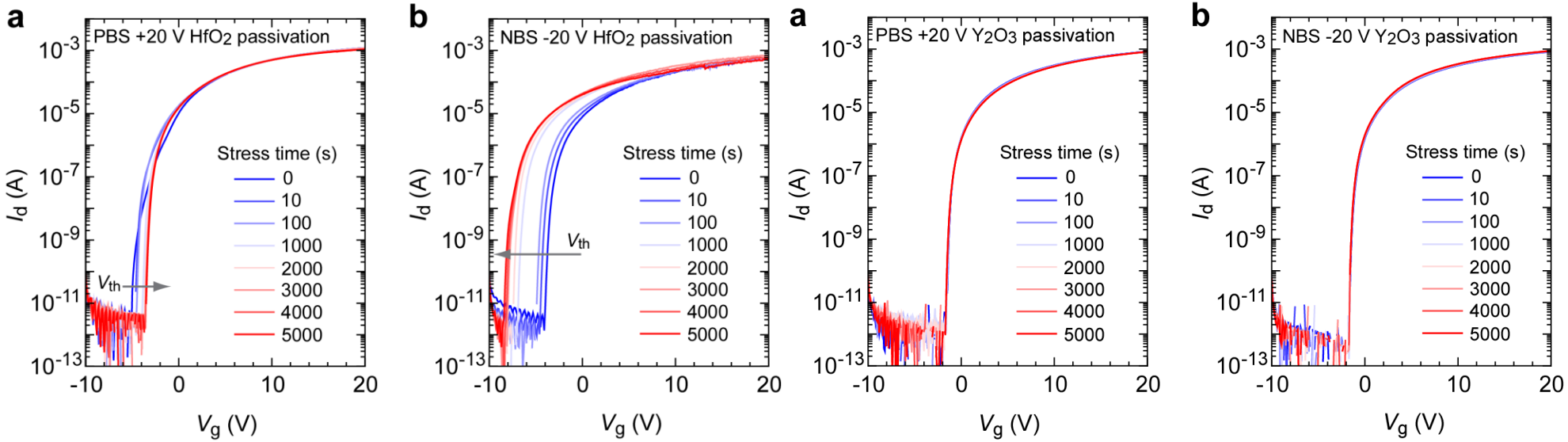
- Before applying bias stress → $\mu_{FE} \approx 85 cm^2V^{-1}s^{-1}$, V_{th} = -3.2V, SS ≈ 0.22V/decade, no hysteresis behavior ⇒ Excellent!
- After applying bias stress → PBS for 5000s: V_{th} +1V shift, NBS for 5000s: V_{th} -10V shift ⇒ Poor…



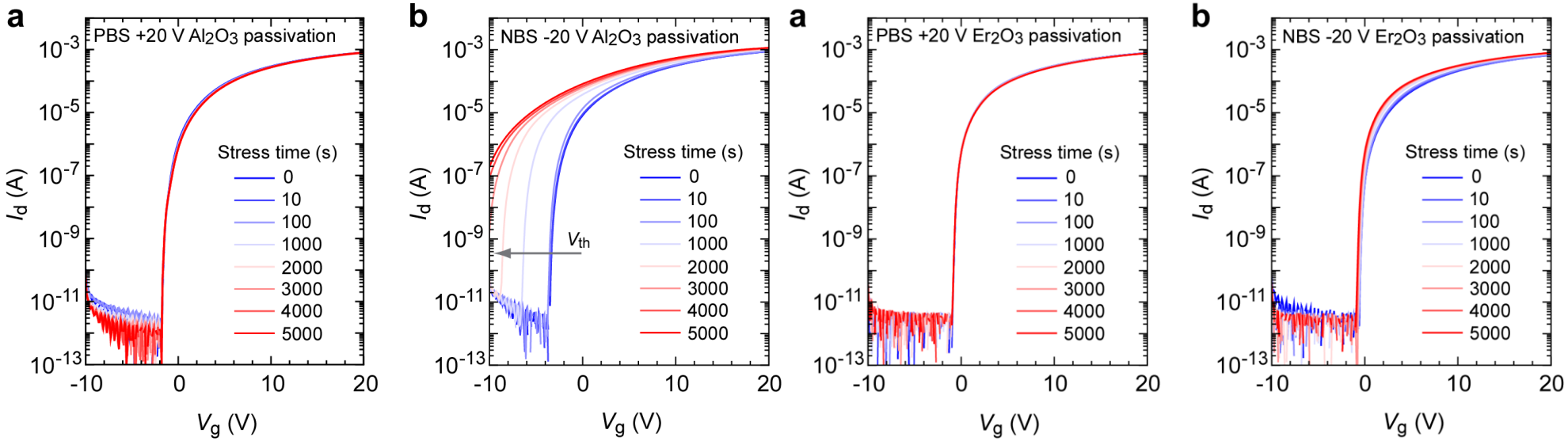
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The In_2O_3 TFTs with various passivation layers

The bias stability of the In_2O_3 TFT passivated with HfO_2, Al_2O_3, Y_2O_3, Er_2O_3 films



Passi.	μ_{FE}	V_{th}	SS
without	93.5	–1.3	0.08
HfO_2	70.6	–4.5	0.18
Al_2O_3	77.8	–1.5	0.1
Y_2O_3	77.6	–4.7	0.16
Er_2O_3	71.5	–0.7	0.09

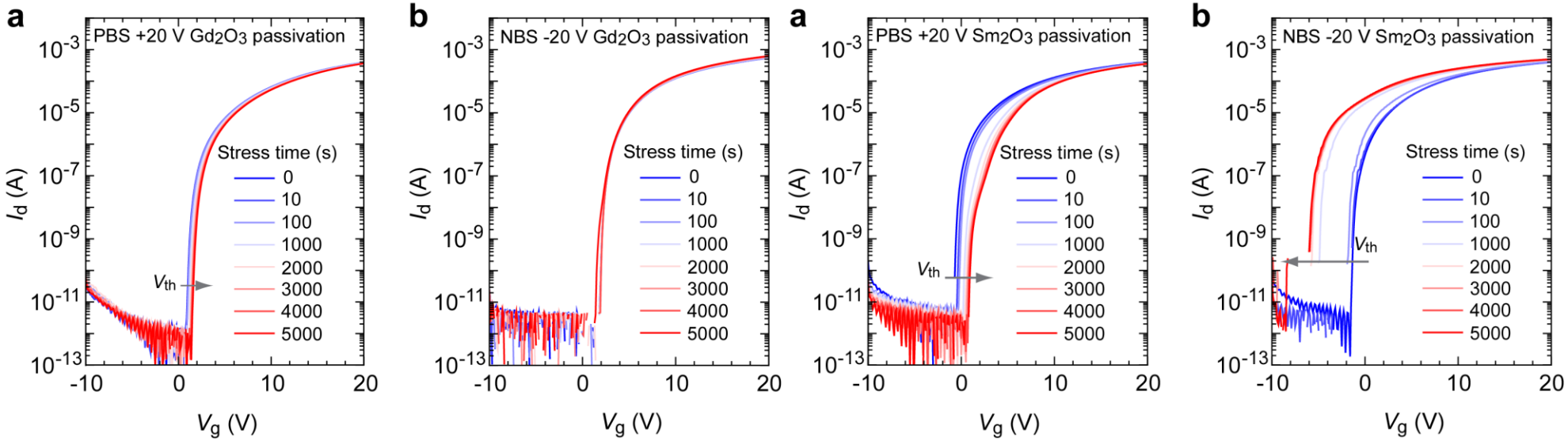


Passi.	PBS	NBS
Without	+1.0V	–10.0V
HfO_2	+1.2V	–4.4V
Al_2O_3	+0.02V	–7.0V
Y_2O_3	+0.1V	–0.1V
Er_2O_3	+0.1V	–0.3V

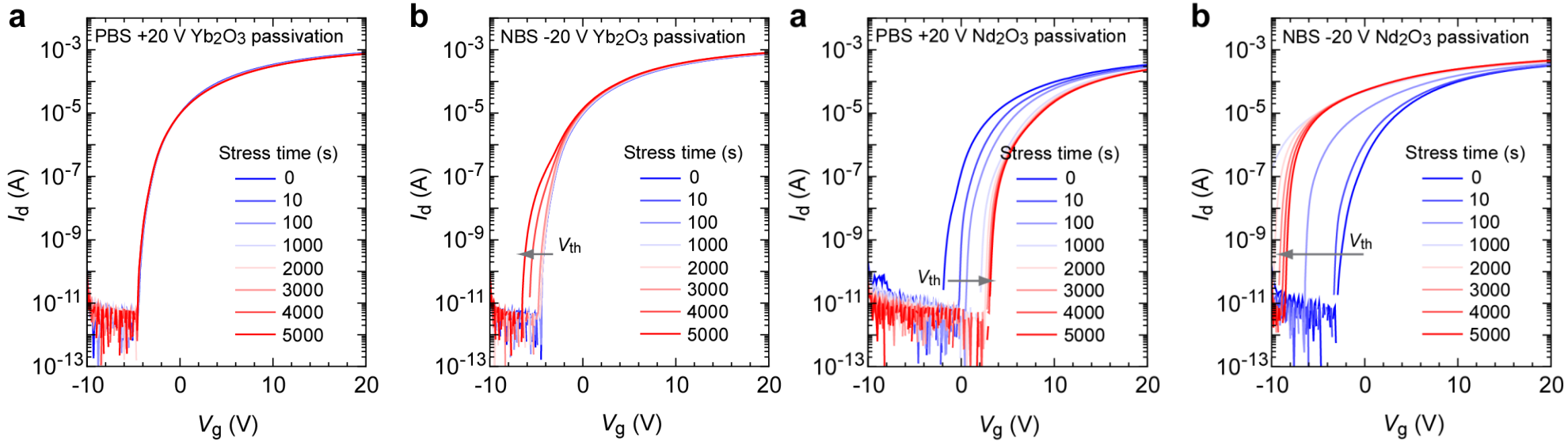
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The In_2O_3 TFTs with various passivation layers

The bias stability of the In_2O_3 TFT passivated with Gd_2O_3, Yb_2O_3, Sm_2O_3, Nd_2O_3 films



Passi.	μ_{FE}	V_{th}	SS
Gd_2O_3	53.6	1.2	0.12
Yb_2O_3	70.0	–4.0	0.16
Sm_2O_3	39.6	–0.6	0.1
Nd_2O_3	40.3	–1.5	0.16



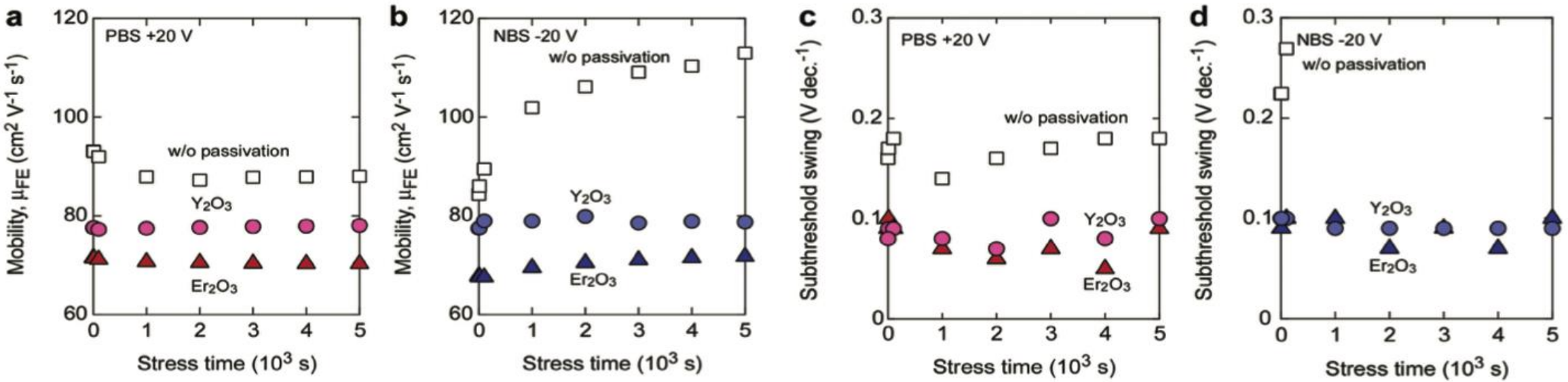
Passi.	PBS	NBS
Gd_2O_3	+0.6V	-0.4V
Yb_2O_3	+0.1V	-1.9V
Sm_2O_3	+1.7V	–4.7V
Nd_2O_3	+4.9V	–6.0V

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Performance degradation of the In_2O_3 TFTs

Reliable device operation in In_2O_3 TFTs passivated with Y_2O_3 and Er_2O_3 films

- the μ_{FE} and SS values of the In_2O_3 TFTs passivated with Y_2O_3 and Er_2O_3 are highly stable against the bias stress application.
- Y_2O_3 passivated TFT $\mu_{FE} \approx 78 cm^2V^{-1}s^{-1}$, Er^2O^3 passivated TFT $\mu_{FE} \approx 70 cm^2V^{-1}s^{-1}$ and both TFT's SS ≈ 0.1V/decade.

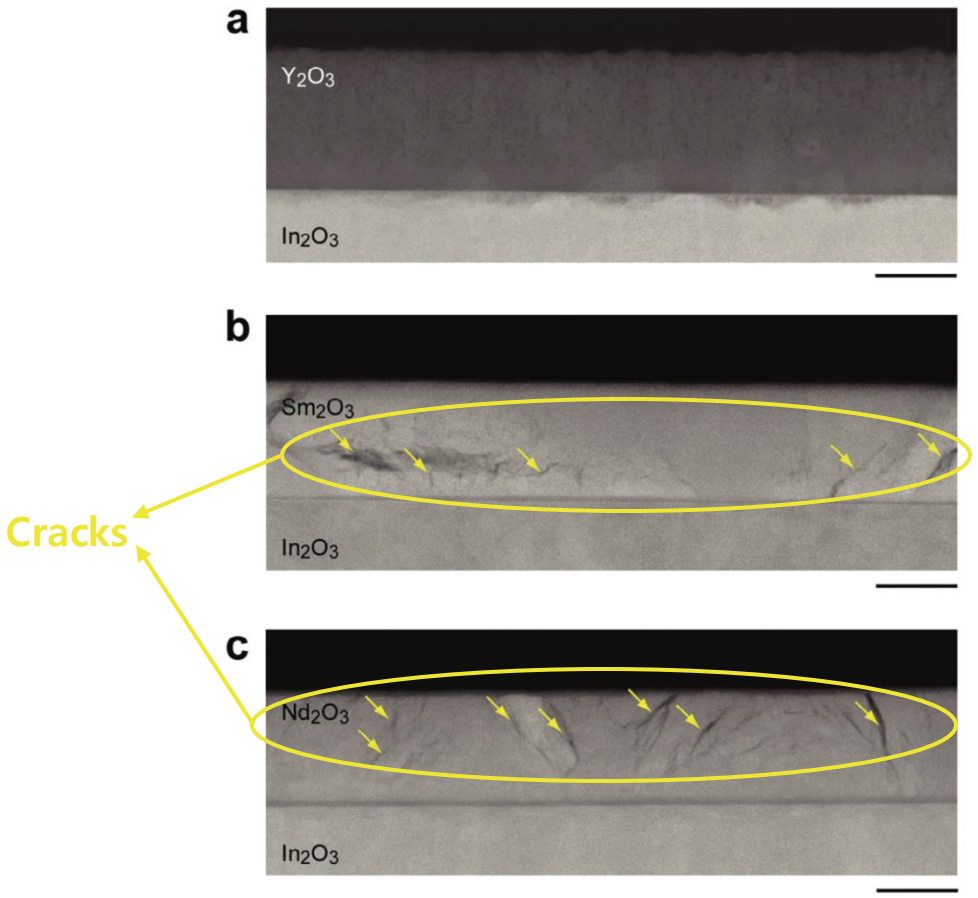


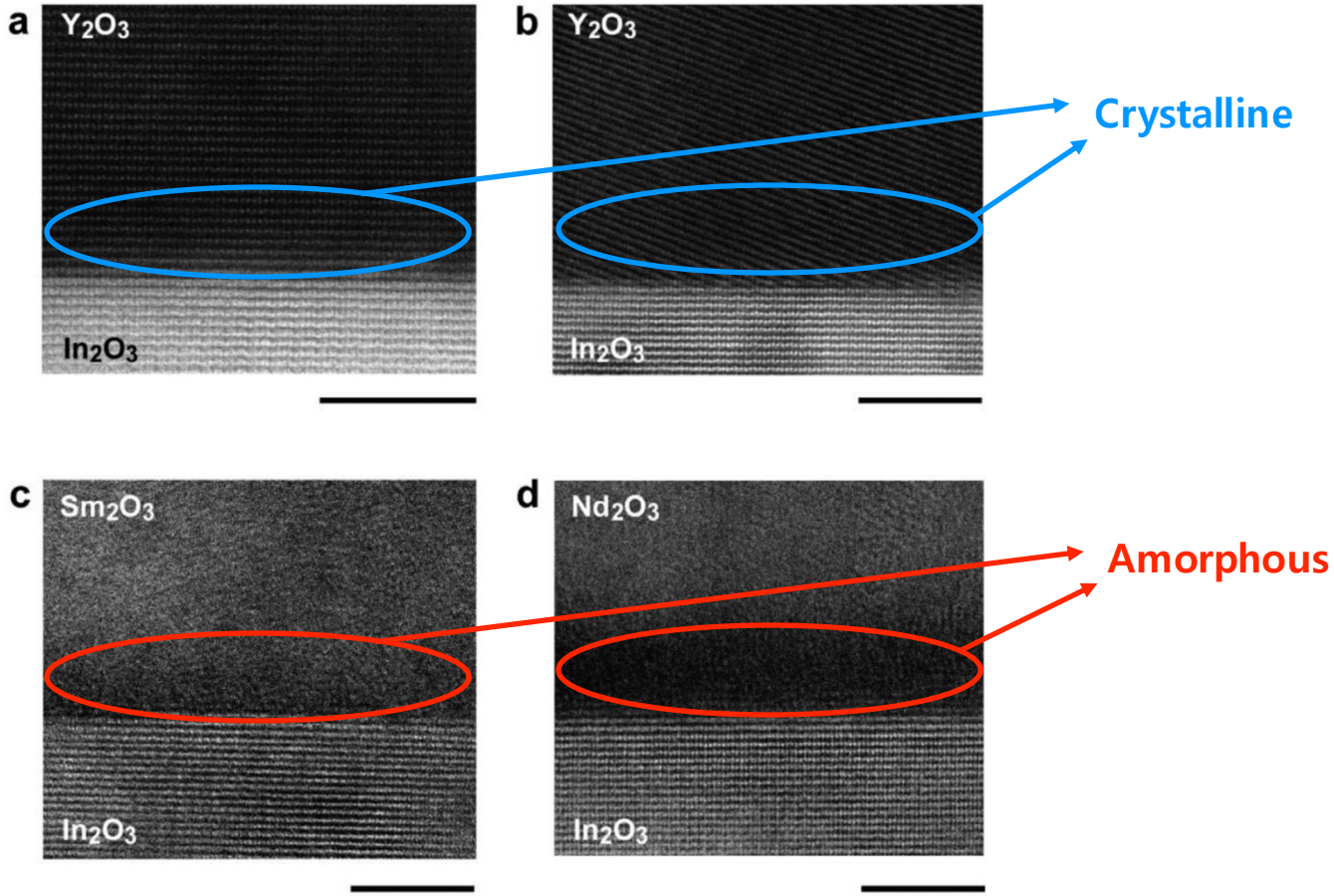
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The origin of the reliability of In_2O_3 TFT with passi.

The heteroepitaxial growth of the passivation materials(Y_2O_3 film)

- Cracks are observed in the Sm_2O_3 and Nd_2O_3 films, such microstructures are not observed in the Y_2O_3 film.
- Lattice structures(heteroepitaxial growth) are observed in the case of Y_2O_3 film along with In_2O_3 lattice

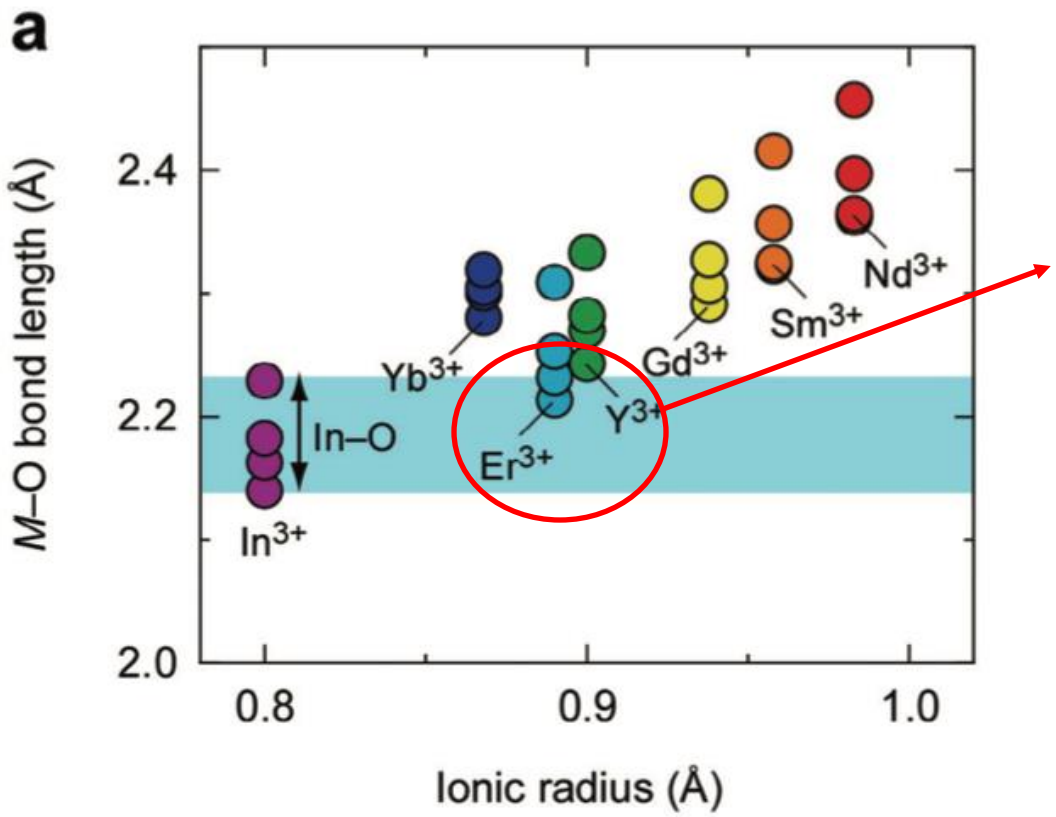




The origin of the reliability of In_2O_3 TFT with passi.

The relationship between the ionic radius and metal–oxygen bond (M–O) length

- Ln_2O_3 has three different crystal structures (i.e., C-, B-, and A-type) depending on the ionic radius.
- The C-type structures (Yb_2O_3, Er_2O_3, Y_2O_3, and Gd_2O_3) are cubic bixbyite and have the same crystal structure as In_2O_3 with an ionic radius of 0.80Å.



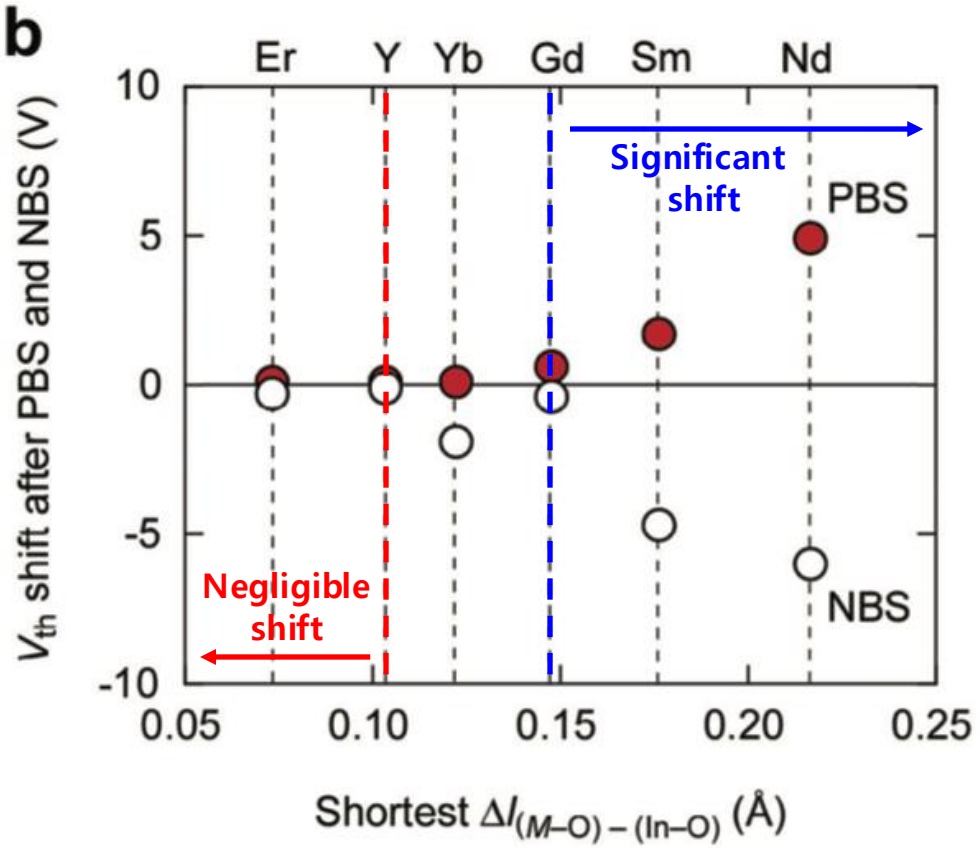
the M–O lengths of Er–O and Y–O are close to In–O.
Therefore, heteroepitaxial growth of Y_2O_3 occurred on In_2O_3.

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The origin of the reliability of In_2O_3 TFT with passi.

The V_{th} shifts about the difference of the M-O length from In-O length($\Delta l_{(M-O)-(In-O)}$)

- The V_{th} shifts are significant for bias stress when the shortest $\Delta l_{(M-O)-(In-O)}$ of the passivation layer exceeds 0.15 Å(Gd_2O_3).
- The V_{th} shifts are negligible when the shortest $\Delta l_{(M-O)-(In-O)}$ is below 0.10 Å(Y_2O_3 and Er_2O_3).

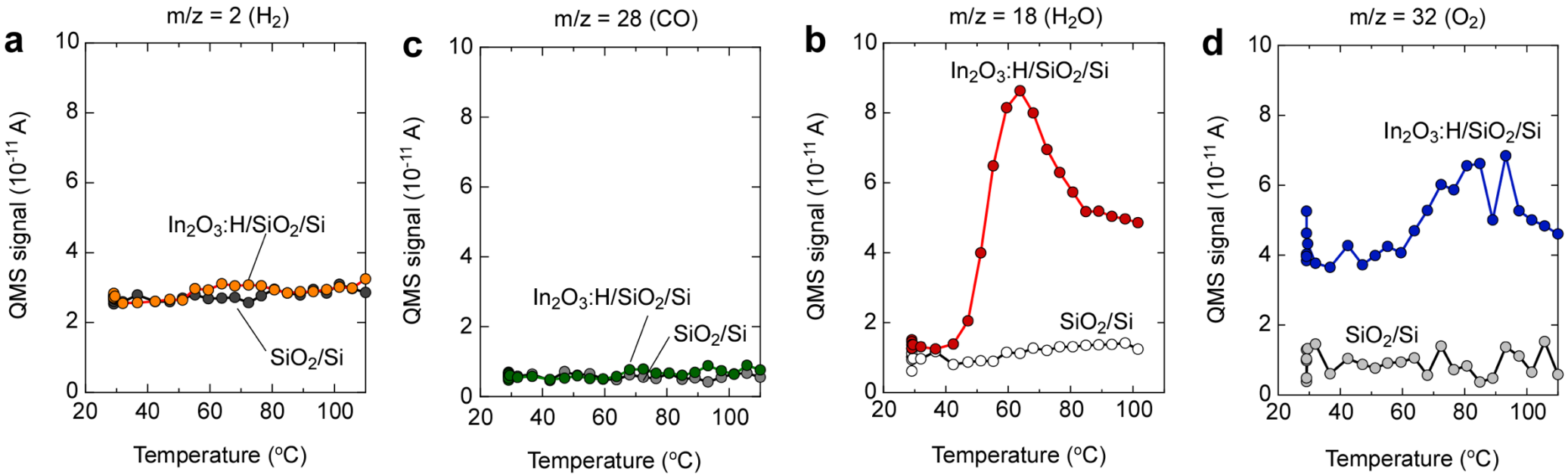


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The origin of the reliability of In_2O_3 TFT with passi.

Thermal desorption spectroscopy(TDS) measurements of the In_2O_3 films

- The H_2 and CO desorption are negligible while small desorption of O_2 and large desorption of H_2O occur from In_2O_3 film.
- The heteroepitaxially grown Y_2O_3 and Er_2O_3 passivation layers would effectively prevent the desorption of OH^- ions from the In_2O_3 films.



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